STEPHEN GREGORY OTIENO

SOFTWARE ENGINEER | FULL-STACK DEVELOPER | AI-ENABLED APPLICATIONS

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Portfolio: Siviq • NexaGen • Top Heights Electricals

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PROFESSIONAL PROFILE

Software developer and digital product builder with an interest in full-stack application development, database-driven systems, mobile experiences, and AI-enabled solutions. Focused on building practical software that solves real problems, with project experience spanning a campus social platform and marketplace, school-management software, and a business website. Brings a product-minded approach to planning, implementation, testing, and continuous improvement.

TECHNICAL SKILLS

Programming languages: JavaScript, TypeScript, Python, PHP, SQL, Java, C++, C#, C, Dart, Kotlin.

Frontend development: HTML5, CSS3, React.js, Next.js, Bootstrap, Tailwind CSS, responsive design, reusable components.

Backend development: Node.js, Express.js, Django, Flask, FastAPI, Laravel, Spring Boot, ASP.NET Core, REST APIs, JSON, webhooks, WebSockets, Socket.IO.

Databases: MySQL, PostgreSQL, SQLite, SQL Server, MongoDB, Firebase Firestore, Redis; database design, queries, indexing, and transactions.

Mobile development: Flutter, Dart, Kotlin, React Native, Firebase services.

Object-oriented programming and design: Encapsulation, inheritance, polymorphism, abstraction, interfaces, composition, SOLID principles, MVC, Repository, Factory, Strategy, and Observer patterns.

AI and data tools: ChatGPT, OpenAI APIs, Google Gemini, GitHub Copilot, Claude, Jupyter, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, LangChain, LlamaIndex, retrieval-augmented generation (RAG), embeddings, and vector search.

Development and deployment tools: Git, GitHub, GitLab, VS Code, IntelliJ IDEA, PyCharm, Android Studio, Postman, Docker, GitHub Actions, Linux, Bash, AWS, Azure, Google Cloud, Vercel, Netlify, Render, and Railway.

Testing and security: Jest, Vitest, Pytest, JUnit, PHPUnit, Playwright, Cypress, Swagger/OpenAPI, input validation, secure configuration, debugging, and awareness of common web vulnerabilities.

EDUCATION

Bachelor of Science in Software Engineering | Kirinyaga University

Bachelor of Science in Computer Science | University of the People, Online

SELECTED PROJECTS

SIVIQ | Campus Social Platform and Marketplace

Website: siviq.top

- Built a campus-focused social platform concept combining posts, messaging, real-time interactions, and an integrated marketplace.
- Worked on bringing student connection and buying-and-selling experiences into one product, with attention to usability and campus community needs.

NEXAGEN | School Management Software

Website: nexagen.co.ke

- Developed a school-management software solution focused on organizing school information and supporting digital administrative workflows.
- Focused on application interfaces, structured information management, and opportunities to reduce repetitive manual processes.

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SOFTWARE ENGINEERING • DIGITAL PRODUCTS • AI

SELECTED PROJECTS — CONTINUED

TOP HEIGHTS ELECTRICALS | Business Website

Website: topheightselectricals.com

- Worked on a business website presenting electrical, solar, and backup-power solutions to online visitors.
- Contributed to a responsive online presence, clear service information, and customer enquiry pathways.

AI-ASSISTED DEVELOPMENT

Uses AI tools to support coding research, explore implementation approaches, explain unfamiliar code, draft and refactor solutions, brainstorm test cases, debug issues, and improve technical documentation. Treats generated output as a starting point that must be reviewed, tested, and adapted to the project.

AI assistants and APIs: ChatGPT, OpenAI API, Google Gemini, GitHub Copilot, and Claude.

AI application concepts and libraries: Prompt design, structured outputs, embeddings, retrieval-augmented generation (RAG), LangChain, LlamaIndex, FAISS, Chroma, Pinecone, pandas, NumPy, scikit-learn, PyTorch, and TensorFlow.

Application patterns: Conversational interfaces, document and knowledge search, intelligent information retrieval, and workflow automation.

ENGINEERING PRACTICES

- Apply modular design, separation of concerns, object-oriented principles, and maintainable coding practices.
- Work with requirements, technical documentation, API design, database modelling, debugging, and version control.
- Use unit, integration, and end-to-end testing approaches to improve reliability and catch regressions.
- Understand deployment fundamentals including environment variables, DNS, SSL/TLS, hosting configuration, logs, and application monitoring.
- Apply security fundamentals such as authentication and authorization, password hashing, input validation, least privilege, and secure handling of credentials.
- Use iterative development, task breakdown, and feedback to improve software products.

PROFESSIONAL EXPERIENCE

Web Development and Digital Solutions | Top Heights Electricals

- Contributed to the company's digital presence through website development and customer-facing service information.
- Supported improvements to website presentation, responsiveness, usability, and troubleshooting.
- Worked toward aligning the website experience with business services and customer needs.

CORE STRENGTHS

Problem-solving • Product thinking • Continuous learning • Technical research • Debugging • User-focused development • Collaboration • Adaptability

PROFESSIONAL LINKS

Portfolio projects: Siviq | NexaGen | Top Heights Electricals